

MULTI ANSWER QUESTIONS

55. Let $\vec{a} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.

Then the diagonals of the parallelogram are

$$\vec{p} = 3\hat{i} + 6\hat{j} - 2\hat{k}, \vec{q} = -\hat{i} - 2\hat{j} + 8\hat{k}$$

So, unit vectors along the diagonals are

$$\frac{1}{7}(3\hat{i} + 6\hat{j} - 2\hat{k}) \text{ and } \frac{1}{\sqrt{69}}(-\hat{i} - 2\hat{j} + 8\hat{k})$$

56. We have,

$$\vec{AB} = -\hat{i} - \hat{j} - 4\hat{k}, \vec{BC} = -3\hat{i} + 3\hat{j} \text{ and}$$

$$\vec{CA} = 4\hat{i} - 2\hat{j} + 4\hat{k}$$

$$\text{Therefore, } |\vec{AB}| = |\vec{BC}| = 3\sqrt{2} \text{ and } |\vec{CA}| = 6$$

$$\text{Clearly, } |\vec{AB}|^2 + |\vec{BC}|^2 = |\vec{AC}|^2$$

Hence, the triangle is right-angled isosceles triangle

57. use $[\vec{a} \vec{b} \vec{c}] = 0$

$$58. \frac{x}{1} = \frac{-2}{y} = \frac{5}{-z} = k \text{ (say)}$$

where $k \in R$

59. $\vec{a} = 2\vec{i} - 2\vec{j} + \vec{k}, \vec{b} = \vec{i} + 2\vec{j} - 2\vec{k}$

$$\therefore |\vec{a}| = 3, |\vec{b}| = 3$$

$\therefore$ a vector along bisectors

$$= \frac{\vec{a}}{|\vec{a}|} \pm \frac{\vec{b}}{|\vec{b}|} = \frac{2\vec{i} - 2\vec{j} + \vec{k}}{3} \pm \frac{\vec{i} + 2\vec{j} - 2\vec{k}}{3}$$

$$= \vec{i} - \frac{1}{3}\vec{k}, \frac{1}{3}\vec{i} - \frac{4}{3}\vec{j} + \vec{k}$$

$\therefore$ the required vector = 2

$$= \frac{\vec{i} - \frac{1}{3}\vec{k}}{\sqrt{1^2 + \left(-\frac{1}{3}\right)^2}} + 2 \cdot \frac{\frac{1}{3}\vec{i} - \frac{4}{3}\vec{j} + \vec{k}}{\sqrt{\left(\frac{1}{3}\right)^2 + \left(-\frac{4}{3}\right)^2 + 1^2}}$$

60. $\vec{OA} = -4\hat{i} + 3\hat{k}; \vec{OB} = 14\hat{i} + 2\hat{j} - 5\hat{k}$

$$\hat{a} = \frac{-4\hat{i} + 3\hat{k}}{5}; \hat{b} = \frac{14\hat{i} + 2\hat{j} - 5\hat{k}}{15}$$

$$\vec{r} = \frac{\lambda}{15}[-12\hat{i} + 9\hat{j} + 14\hat{i} + 2\hat{j} - 5\hat{k}]$$

$$\vec{r} = \frac{\lambda}{15}[2\hat{i} + 2\hat{j} + 5\hat{k}]$$

$$\vec{r} = \frac{2\lambda}{15}[\hat{i} + \hat{j} + 2\hat{k}]$$

61. For coplanar vectors, $\begin{vmatrix} 1 & 2 & 3 \\ 0 & \lambda & \mu \\ 0 & 0 & 2\lambda - 1 \end{vmatrix} = 0$

$$\Rightarrow (2\lambda - 1)\lambda = 0 \Rightarrow \lambda = 0, \frac{1}{2}$$

62. Let $\vec{\alpha} = \hat{i} + x\hat{j} + 3\hat{k}, \vec{\beta} = 4\hat{i} + (4x - 2)\hat{j} + 2\hat{k}$

$$\text{Given, } 2|\vec{\alpha}| = |\vec{\beta}|$$

$$\Rightarrow 2\sqrt{10 + x^2} = \sqrt{20 + 4(2x - 1)^2}$$

$$\Rightarrow 10 + x^2 = 5 + (4x^2 - 4x + 1)$$

$$\Rightarrow 3x^2 - 4x - 4 = 0$$

$$\Rightarrow x = 2, -\frac{2}{3}$$

63. We have, $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2(\vec{a} \cdot \vec{b})$

$$\Rightarrow |\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos 2\theta$$

$$\Rightarrow |\vec{a} - \vec{b}|^2 = 2 - 2\cos 2\theta \left(\because |\vec{a}| = |\vec{b}| = 1 \right)$$

$$\Rightarrow |\vec{a} - \vec{b}|^2 = 4\sin^2 \theta$$

$$\Rightarrow |\vec{a} - \vec{b}| = 2|\sin \theta|$$

$$\text{Now, } |\vec{a} - \vec{b}| < 1$$

$$\Rightarrow 2|\sin \theta| < 1$$

$$\Rightarrow |\sin \theta| < \frac{1}{2}$$

$$\Rightarrow \theta \in \left[0, \frac{\pi}{6}\right] \text{ or } \theta \in \left(\frac{5\pi}{6}, \pi\right)$$

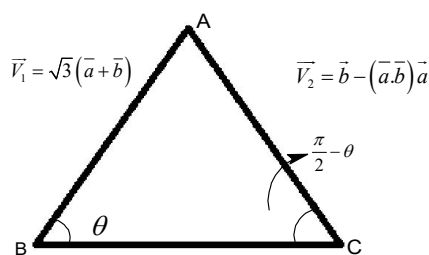
$$64. \quad |\vec{a} + \vec{b}| = |\vec{a} - 2\vec{b}|$$

$$\Rightarrow \vec{a} \cdot \vec{b} = \frac{|\vec{b}|^2}{2}$$

$$\text{Also } \vec{a} \cdot \vec{b} + \frac{1}{|\vec{b}|^2 + 2} = \frac{|\vec{b}|^2 + 2}{2} + \frac{1}{|\vec{b}|^2 + 2} - 1$$

$$\geq \sqrt{2} - 1 \text{ [Using A.M.} \geq \text{G.M.]}$$

$$65. \quad \text{Consider } \vec{V}_1 \cdot \vec{V}_2 = 0 \Rightarrow A = 90^\circ$$



$$\text{Using the sine law, } \frac{|\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}|}{\sin \theta} = \frac{\sqrt{3}|\vec{a} \times \vec{b}|}{\cos \theta}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}} \frac{|\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}|}{|\vec{a} \times \vec{b}|} = \frac{1}{\sqrt{3}} \frac{|(\vec{a} \times \vec{b}) \times \vec{a}|}{|\vec{a} \times \vec{b}|}$$

$$= \frac{1}{\sqrt{3}} \frac{|(\vec{a} \times \vec{b})||\vec{a}|\sin 90^\circ}{|\vec{a} \times \vec{b}|} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$66. \quad \text{We know that } \vec{a} + \vec{b} + \vec{c} = \vec{0}, \text{ then}$$

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

$$\text{Given } \vec{a} + 2\vec{b} + 3\vec{c} = \vec{0}$$

$$\Rightarrow 2\vec{a} \times \vec{b} = 6\vec{b} \times \vec{c} = 3\vec{c} \times \vec{a}$$

$$\text{Hence, } \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = 2(\vec{a} \times \vec{b})$$

$$\text{or } 6(\vec{b} \times \vec{c}) \text{ or } 3(\vec{c} \times \vec{a})$$

$$67. \quad \vec{a} \times \vec{b} = \vec{c}, \vec{b} \times \vec{c} = \vec{a}$$

Taking cross with $\vec{b}$ in first equation, we get

$$\vec{b} \times (\vec{a} \times \vec{b}) = \vec{b} \times \vec{c} = \vec{a}$$

$$\Rightarrow |\vec{b}|^2 \vec{a} - (\vec{a} \cdot \vec{b})\vec{b} = \vec{a} \Rightarrow |\vec{b}| = 1 \text{ and } \vec{a} \cdot \vec{b} = 0$$

$$\text{Also } |\vec{a} \times \vec{b}| = |\vec{c}| \Rightarrow |\vec{a}||\vec{b}|\sin \frac{\pi}{2} = |\vec{c}|$$

$$\Rightarrow |\vec{a}| = |\vec{c}|$$

$$68. \quad \vec{x} \cdot \vec{y} = \vec{y} \cdot \vec{z} = \vec{z} \cdot \vec{x} = \sqrt{2} \times \sqrt{2} \times \frac{1}{2} = 1.$$

$$\text{Let } \vec{a} = \lambda(\vec{x} \times (\vec{y} \times \vec{z})) = \lambda((\vec{x} \cdot \vec{z})\vec{y} - (\vec{x} \cdot \vec{y})\vec{z})$$

$$= \lambda(\vec{y} - \vec{z})$$

$$\vec{a} \cdot \vec{y} = \lambda$$

$$\therefore \vec{a} = (\vec{a} \cdot \vec{y})(\vec{y} - \vec{z})$$

$$\text{Similarly } \vec{b} = (\vec{b} \cdot \vec{z})(\vec{z} - \vec{x})$$

$$\vec{a} \cdot \vec{b} = -(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z})$$

$$69. \quad \text{The given vector equation can be written as}$$

$$[(1-a)x + 3y - 4z]\hat{i} + [x - (3+a)y + 5z]\hat{j}$$

$$+ [3x + y - az]\hat{k} = \vec{0}$$

$$\Rightarrow (1-a)x + 3y - 4z = 0$$

$$x - (3+a)y + 5z = 0 \text{ and } 3x + y - az = 0$$

The above system of equation has a non-trivial solution

$$\Rightarrow \begin{vmatrix} 1-a & 3 & -4 \\ 1 & -(3+a) & 5 \\ 3 & 1 & -a \end{vmatrix} = 0$$

$$\Rightarrow a = 0, -1$$

$$70. \quad \text{We have } \vec{a} = 2p\hat{i} + \hat{j}$$

on rotation, let $\vec{b}$ the vector with components

$(p+1)$ and 1 so that $\vec{b} = (p+1)\hat{i} + \hat{j}$

Now, $|\vec{a}| = |\vec{b}| \Rightarrow a^2 = b^2$

$$\Rightarrow 4p^2 + 1 = (p+1)^2 + 1$$

$$\Rightarrow 4p^2 = (p+1)^2 + 1$$

$$\Rightarrow 2p = \pm(p+1) \Rightarrow 3p = -1 \text{ or } p = 1$$

$$\therefore p = -1/3 \text{ or } p = 1$$

$$71. \quad \vec{u} \cdot \vec{a} = |\vec{a}|^2 - |\vec{a} \cdot \vec{b}|^2$$

$$|\vec{v}|^2 = |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$$

$$|\vec{u}|^2 = |\vec{a}|^2 |\vec{b}|^2 + (\vec{a} \cdot \vec{b}) |\vec{b}|^2 - 2(\vec{a} \cdot \vec{b})^2$$

$$= |\vec{a}|^2 + |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$$

$$72. \quad \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} \text{ and}$$

$$(\vec{a} \times \vec{b}) \times \vec{c} = -(\vec{c} \cdot \vec{b}) \vec{a} + (\vec{a} \cdot \vec{c}) \vec{b}$$

We have been given

$$(\vec{a} \times (\vec{b} \times \vec{c})) \cdot ((\vec{a} \times \vec{b}) \times \vec{c}) = 0.$$

Therefore,

$$((\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}) \cdot ((\vec{a} \cdot \vec{c}) \vec{b} - (\vec{c} \cdot \vec{b}) \vec{a}) = 0$$

$$(\vec{a} \cdot \vec{c})^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{c})(\vec{a} \cdot \vec{b})$$

$$- (\vec{a} \cdot \vec{b})(\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{c}) + (\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{c})(\vec{c} \cdot \vec{a}) = 0$$

$$(\vec{a} \cdot \vec{c})^2 |\vec{b}|^2$$

$$= (\vec{a} \cdot \vec{c})(\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{c})(\vec{a} \cdot \vec{c})((\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{b}) - (\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{c}))$$

$$= 0$$

$$\vec{a} \cdot \vec{c} = 0 \text{ or } (\vec{a} \cdot \vec{c}) |\vec{b}|^2 = (\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{c})$$

$$73. \quad \text{Let } \vec{\alpha} = \hat{i} - \hat{j} - \hat{k}, \vec{\beta} = \hat{i} + \hat{j} + \hat{k} \text{ and}$$

$$\vec{\gamma} = -\hat{i} + \hat{j} + \hat{k}$$

$$\text{Let } \vec{a} = \hat{i} + \hat{k}, \vec{b} = 2\hat{i} + \hat{j} + \hat{k},$$

$$\vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}, \vec{d} = -4\hat{i} - 2\hat{j} - 2\hat{k}$$

Clearly, $\vec{\alpha}$ is perpendicular to $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$

$$\text{Also, } [\vec{\alpha} \vec{\beta} \vec{\gamma}] = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{vmatrix} = 2 \neq 0$$

$$[\vec{b} \vec{\beta} \vec{\gamma}] = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{vmatrix} = 0$$

$$[\vec{c} \vec{\beta} \vec{\gamma}] = \begin{vmatrix} 3 & 2 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{vmatrix} = 2 \neq 0$$

$$[\vec{d} \vec{\beta} \vec{\gamma}] = \begin{vmatrix} -4 & -2 & -2 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{vmatrix} = 0$$

74. It is given that $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$ are coplanar vectors. Therefore,

$$[\vec{\alpha} \vec{\beta} \vec{\gamma}] = 0$$

$$\Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$$\Rightarrow 3abc - a^3 - b^3 - c^3 = 0$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = 0$$

$$\Rightarrow (a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 0$$

$$\Rightarrow a+b+c = 0 \left[\because a^2+b^2+c^2-ab-bc-ca \neq 0 \right]$$

$$\Rightarrow \vec{v} \cdot \vec{\alpha} = \vec{v} \cdot \vec{\beta} = \vec{v} \cdot \vec{\gamma} = 0$$

$$\Rightarrow \vec{v} \text{ is perpendicular to } \vec{\alpha}, \vec{\beta} \text{ and } \vec{\gamma}$$

75. For $\vec{A}, \vec{B}$ and $\vec{C}$ to form a left-handed system

$$[\vec{A} \vec{B} \vec{C}] < 0$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & 1 & 5 \end{vmatrix} = 11\hat{i} - 6\hat{j} - \hat{k}$$

(1) is satisfied by options (B) and (D)

76. $\therefore 2\hat{i} + 3\hat{j} - 3\hat{k}$ & $-2\hat{i} - \hat{j} + 5\hat{k}$ satisfies both conditions.

77. Clearly (B), (D) are meaningless

78. Let vector $\vec{AO}$ be parallel to line of intersection of planes P_1 and P_2 through origin.

$$\text{i.e. } [(2\hat{j} + 3\hat{k}) \times (4\hat{j} - 3\hat{k})] \times [(j - k) \times (3i + 3j)]$$

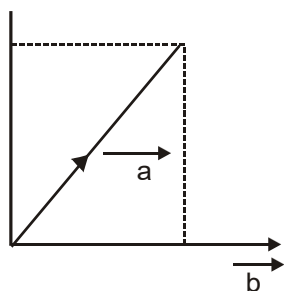
$$79. \vec{x} \times \vec{b} = \vec{c} \times \vec{b} \Rightarrow \vec{a} \times (\vec{x} \times \vec{b}) = \vec{a} \times (\vec{c} \times \vec{b})$$

$$\text{or } \vec{x} = \frac{\vec{a} \times (\vec{c} \times \vec{b})}{\vec{a} \cdot \vec{b}} (\because \vec{a} \perp \vec{x})$$

$$\text{Also, } \vec{a} \times (\vec{c} \times \vec{b}) = -(\vec{c} \times \vec{b}) \times \vec{a} = (\vec{b} \times \vec{c}) \times \vec{a}$$

80. The magnitude of the resolved part along the

$$\text{vector } \vec{b} = \vec{a} \cdot \frac{\vec{b}}{|\vec{b}|}$$



$\therefore$ the resolved part along the vector $\vec{b}$

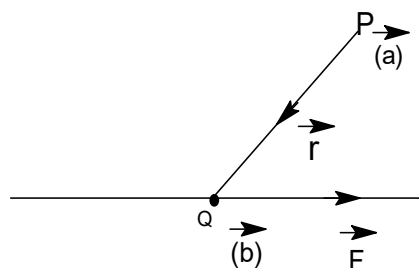
$$= \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \right) \frac{\vec{b}}{|\vec{b}|} = \frac{(\vec{a} \cdot \vec{b}) \vec{b}}{b^2}$$

$\therefore$ resolved part perpendicular to

$$\vec{b} = \vec{a} - \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \right) \frac{\vec{b}}{|\vec{b}|} = \frac{(\vec{b} \cdot \vec{b}) \vec{a} - (\vec{a} \cdot \vec{b}) \vec{b}}{b^2} = \frac{\vec{b} \times (\vec{a} \times \vec{b})}{b^2}$$

STATEMENT QUESTIONS

81. Then, moment of force $\vec{F}$ about point $P(\vec{a})$ is $\vec{r} \times \vec{f}$ when $\vec{r} = \vec{PQ}$



$$82. 2\vec{a} + 3\vec{b} - 5\vec{c} = \vec{0}$$

$$\Rightarrow 3(\vec{b} - \vec{a}) = 5(\vec{c} - \vec{a}) \Rightarrow \vec{AB} = \frac{5}{3} \vec{AC}$$

83. In an isosceles triangle ABC in which $AB=AC$, the median and the bisector from A must be the same line $\Rightarrow$ statement 2 is true.

Now

$$\vec{AD} = \frac{\vec{u} + \vec{v}}{2} \quad \text{and} \quad |\vec{AD}|^2 = \frac{1}{2} 2 \cos^2 \frac{\alpha}{2}; \quad \text{so}$$

$$|\vec{AD}| = \cos \frac{\alpha}{2}$$

Therefore, unit vector along AD, i.e., $\hat{x}$, is given

$$\text{by } \hat{x} = \frac{\vec{AD}}{|\vec{AD}|} = \frac{\vec{u} + \vec{v}}{2 \cos \frac{\alpha}{2}}$$

84. Obviously, statement 1 is true

$$\cos 2\alpha + \cos 2\beta + \cos 2\gamma$$

$$= 2 \cos^2 \alpha - 1 + 2 \cos^2 \beta - 1 + 2 \cos^2 \gamma - 1$$

85. conceptual

$$86. \frac{1}{2} = \frac{p}{3} = \frac{2}{q} \Rightarrow p = \frac{3}{2}, q = 4$$

87. $\vec{a}, \vec{b}, \vec{c}$ are coplanar $\vec{c} = \lambda \vec{a} + \mu \vec{b} \Rightarrow \lambda = 4$ and $\mu = -1$

$$88. \quad \vec{OI} = \frac{|\vec{CB}||\vec{OA}| + |\vec{CA}||\vec{OB}| + |\vec{AB}||\vec{OC}|}{|\vec{BC}| + |\vec{CA}| + |\vec{AB}|}$$

PASSAGE QUESTIONS

$$89. \quad \text{Vector along internal angular bisector} = \frac{\vec{b}}{|\vec{b}|} + \frac{\vec{c}}{|\vec{c}|}$$

$$= \frac{1}{9} (i - 7j + 2k)$$

$$\therefore \vec{d} = 5\sqrt{6} \left(\frac{i - 7j + 2k}{3\sqrt{6}} \right) = \frac{5}{3} (i - 7j + 2k)$$

$$90. \quad BD, BC, BE \text{ are in } H.P$$

$$\Rightarrow \frac{2}{BC} = \frac{1}{BD} + \frac{1}{BE}$$

$$91. \quad \vec{OI}_3 = \frac{-a\vec{a} + b\vec{b} + c\vec{c}}{-a + b + c}$$

$$92. \quad \vec{OA} + \vec{OB} + \vec{OC} = 2\vec{OO}$$

$$93. \quad \vec{NA} + \vec{NB} + \vec{NC} = (\vec{OA} + \vec{OB} + \vec{OC}) - 3\vec{ON}$$

$$94. \quad \vec{SA} + \vec{SB} + \vec{SC} = \vec{GA} + \vec{GB} + \vec{GC} - 3\vec{GS} \\ = \vec{O} - 3\vec{GS} \\ = 3\vec{SG}$$

$$95. \quad p > 0, q > 0, p + q < 1$$

$$96. \quad \vec{c} = \vec{a} + \frac{4}{3}\vec{b}$$

$$\Rightarrow p > 0, q > 0, p + q > 1$$

$$97. \quad p < 0, q > 0$$

$$98. \quad p > 0, q < 0$$

$$99. \quad \text{The angle between the line and the plane is}$$

$$\frac{\pi}{2} - \cos^{-1} \left| \frac{1 \times 1 - 1 \times 1 - 1 \times 1}{\sqrt{3} \times \sqrt{3}} \right| = \sin^{-1} \left(\frac{1}{3} \right)$$

$$100. \quad \text{Solving the two equations}$$

$$\text{We get } \vec{r} = \hat{i}$$

$$101. \quad \text{The vector normal to the plane is } \hat{i} - \hat{j} - \hat{k}$$

$$\frac{1}{\sqrt{3}} |(\hat{r} - \hat{i}) \cdot (\hat{i} - \hat{j} - \hat{k})| = \frac{1}{\sqrt{3}}$$

$$|t| = 1 \Rightarrow t = \pm 1$$

$$\text{So, the point on the line is } i \pm 1(\hat{i} + \hat{j} + \hat{k})$$

$$\text{i.e. } 2\hat{i} + \hat{j} + \hat{k} \text{ or } -\hat{j} - \hat{k}$$

$$102. \quad \text{If } \theta = \frac{\pi}{2} \text{ then } \vec{\alpha} \times \vec{\beta} = |\vec{\alpha}| |\vec{\beta}|$$

$$103. \quad \vec{\alpha} * \vec{\alpha} = |\vec{\alpha}| |\vec{\alpha}| \tan \left(\frac{0}{2} \right)$$

$$104. \quad \vec{\alpha} * \vec{\beta} = \vec{\alpha} \cdot \vec{\beta}$$

$$\Rightarrow \cos \theta = \tan \frac{\theta}{2} \Rightarrow \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} = \tan \frac{\theta}{2}$$

$$\text{then let } t = \tan \frac{\theta}{2}$$

$$105. \quad \frac{BD}{DC} = \frac{AB}{AC} = \frac{\gamma}{\beta}$$

$$\text{position vector of } D = \frac{\gamma \vec{c} + \beta \vec{b}}{\gamma + \beta}$$

$$\text{Now } BD = \frac{\alpha}{\gamma + \beta} \cdot \gamma$$

$$\text{In } \triangle BDA, BI \text{ is bisector of } \angle BDA \text{ also}$$

$$\frac{ID}{IA} = \frac{BD}{AB} = \frac{\frac{\gamma \alpha}{\gamma + \beta}}{\gamma} = \frac{\alpha}{\gamma + \beta}$$

$$I = \frac{\alpha \vec{a} + \frac{(\gamma + \beta)(\gamma \vec{c} + \beta \vec{b})}{\gamma + \beta}}{\alpha + \beta + \gamma} = \frac{\alpha \vec{a} + \beta \vec{b} + \gamma \vec{c}}{\alpha + \beta + \gamma}$$

$$106. \quad \vec{IB} \cdot \vec{IC} = |\vec{IB}| |\vec{IC}| \cos \left(180^\circ - \left(\frac{B}{2} + \frac{C}{2} \right) \right) \\ = r \cos ec \left(\frac{B}{2} \right) r \cos ec \left(\frac{C}{2} \right) \left(-\cos \left(\frac{B+C}{2} \right) \right)$$

$$= -r^2 \operatorname{cosec}\left(\frac{B}{2}\right) r \operatorname{cosec}\left(\frac{C}{2}\right) \sin\left(\frac{A}{2}\right)$$

$$107. |\vec{IB} \times \vec{IC}| = r^2 \operatorname{cosec}\left(\frac{B}{2}\right) \operatorname{cosec}\left(\frac{C}{2}\right) \cos\left(\frac{A}{2}\right)$$

$$108. V = \frac{1}{6} [\vec{a} \vec{b} \vec{c}] = \frac{1}{6} \sqrt{\begin{vmatrix} 1 & 1/2 & 1/2 \\ 1/2 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{vmatrix}} = \frac{1}{6\sqrt{2}}$$

109. The face DAB has normal along $\vec{a} \times \vec{b}$

The face DBC has normal along $\vec{b} \times \vec{c}$

If θ is the angle between the faces

$$|\vec{a} \times \vec{b}| |\vec{b} \times \vec{c}| \cos \theta = |(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c})|$$

$$= \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} \cos \theta = \begin{vmatrix} 1 & 1/2 \\ 2 & 2 \\ 1 & 1/2 \end{vmatrix} = \frac{1}{4}$$

$$\Rightarrow \cos \theta = \frac{1}{3} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{3}\right)$$

110. The equations DA and BC are

$$\vec{r} = s\vec{a}, \vec{r} = \vec{b} + t(\vec{c} - \vec{b})$$

$$(\vec{c} - \vec{b}) \times \vec{a} \times \vec{b} + \vec{c} \times \vec{a}$$

$$|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}|^2 = |\vec{a} \times \vec{b}|^2 + |\vec{c} \times \vec{a}|^2 + 2(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{a})$$

$$= \frac{3}{4} + \frac{3}{4} + 2 \begin{vmatrix} 1/2 & 1/2 \\ 1 & 1/2 \end{vmatrix} = 1$$

$$\text{The distance} = \vec{b} \cdot (\vec{a} \times \vec{b} + \vec{c} \times \vec{a})$$

$$= \vec{b} \cdot (\vec{c} \times \vec{a}) = \frac{1}{\sqrt{2}}$$

$$111. \vec{PQ} // 2\hat{i} + 7\hat{j} - 5\hat{k}$$

$$\Rightarrow \frac{8+3s+3t}{2} = \frac{4-s-2t}{7} = \frac{-8+s-4t}{-5}$$

$$\Rightarrow 23s + 25t = -48, s - 19t = 18$$

$$\Rightarrow s = t = -1 \Rightarrow p \text{ is } 2\hat{i} + 8\hat{j} - 3\hat{k}$$

$$112. t = -1 \Rightarrow Q \text{ is } \hat{j} + 2\hat{k}$$

$$113. P \text{ is } (2, 8, -3) \text{ and } Q \text{ is } (0, 1, 2)$$

$$\therefore PQ = \sqrt{4 + 49 + 25} = \sqrt{78}$$

114 to 116.

$\vec{FP}$ is a vector of magnitude $\sqrt{2}$ and perpendicular to the plane of ABC.

$$\vec{BC} \times \vec{BA} = 7\hat{j} + 7\hat{k}$$

$$\vec{FP} = \pm(\hat{j} + \hat{k})$$

$$\text{Volume of tetrahedron} = \frac{1}{3}(\Delta ABC)(\text{height})$$

$$= \frac{1}{3} \times \frac{7}{\sqrt{2}} \times \sqrt{2} = \frac{7}{3} \text{ cubic units.}$$

$$D = \left(2, \frac{4}{3}, \frac{8}{3}\right), E = \left(\frac{7}{3}, \frac{-1}{3}, \frac{13}{3}\right)$$

F = point of intersection of BE and CD

$$= (2, 0, 4)$$

$$\therefore \vec{OP} = \vec{OF} + \vec{FP} = (2, 0, 4) + (0, 1, 1) = (2, 1, 5)$$

$$\text{Equation of } \vec{AP} = \vec{r} = 3\hat{i} + \hat{j} + 3\hat{k} + t(\hat{i} - 2\hat{k})$$

117 to 119

$$\vec{r} \text{ lies on the circle } x^2 + y^2 + 8x - 10y + 40 = 0$$

$$\therefore P_2 = \min |\vec{r} - (-2\hat{i} + 3\hat{j})| = 2\sqrt{2} - 1$$

$$P_1 = \max |\vec{r} - (-2\hat{i} + 3\hat{j})| = 2\sqrt{2} + 1$$

MATRIX-MATCHING QUESTIONS

$$120. \vec{OA} = \vec{a}, \vec{AB} = \vec{p} \Rightarrow \vec{OB} = \vec{p} + \vec{a}$$

$$\vec{BC} = \vec{q} \Rightarrow \vec{OC} = \vec{p} + \vec{q} + \vec{a}$$

$$\vec{OD} = \vec{a} + \vec{q}$$

$$121. \vec{OD} = \frac{\vec{OB} + \vec{OC}}{2}, \vec{OE} = \frac{\vec{OC} + \vec{OA}}{2}$$

$$\vec{OA} = \frac{\vec{OA} + \vec{OB}}{2}$$

122. a. $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{6}$

$$\Rightarrow \vec{a}^2 + \vec{b}^2 + \vec{c}^2 + 2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a} = 6$$

$$\therefore |\vec{a}| = 1$$

b. $\vec{a}$ is perpendicular $\vec{b} + \vec{c} \Rightarrow \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = 0$

(1)

$\vec{b}$ is perpendicular $\vec{a} + \vec{c} \Rightarrow \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{c} = 0$

(2)

$\vec{c}$ is perpendicular $\vec{a} + \vec{b} \Rightarrow \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} = 0$

(3)

From (1), (2) and (3), we get

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$$

$$\therefore |\vec{a} + \vec{b} + \vec{c}| = 7$$

c. $(\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{b} \cdot \vec{c})(\vec{a} \cdot \vec{d}) = 21$

d. We know that

$$\begin{bmatrix} \vec{a} \times \vec{b} & \vec{b} \times \vec{c} & \vec{c} \times \vec{a} \end{bmatrix} = \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2$$

$$\text{And } \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{vmatrix}$$

$$= 32$$

$$\therefore \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = 4\sqrt{2}$$

123. (A) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = 0$

$$|\vec{a} + \vec{b} + \vec{c}|^2 = 1 + 2 + 3 + 2 \times 0 = 6$$

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{6}$$

(B) Let $|\vec{a}| = |\vec{b}| = |\vec{c}| = k$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \frac{k^2}{2}$$

$$|\vec{a} + \vec{b} + \vec{c}|^2 = 3k^2 + 2 \times \left(\frac{k^2}{2} \times 3 \right)$$

$$\Rightarrow 6k^2 = 6k^2 \Rightarrow k = 1$$

(C) Let $|\vec{b}| = x$ $\vec{a} \cdot \vec{b} = y$

$$|\vec{a} + \vec{b}|^2 = 100$$

$$25 + x^2 + 2y = 100$$

$$|\vec{a} - \vec{b}|^2 = 64$$

$$\Rightarrow 25 + x^2 - 2y = 64$$

$$\text{So, } x = \sqrt{57}$$

(D) $|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 = -2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$ So,

$$\vec{a} \cdot (\vec{b} + \vec{a}) + \vec{b} \cdot (\vec{c} + \vec{b}) + \vec{c} \cdot (\vec{a} + \vec{c})$$

$$= \frac{1}{2}(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2) > 0$$

124. $P \rightarrow \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = 2$

$$\begin{bmatrix} 2\vec{a} \times \vec{b} & 3\vec{b} \times \vec{c} & \vec{c} \times \vec{a} \end{bmatrix} = 6 \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2 = 6 \times 4 = 24$$

$$Q \rightarrow \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = 5$$

$$6 \begin{bmatrix} \vec{a} + \vec{b} & \vec{b} + \vec{c} & \vec{c} + \vec{a} \end{bmatrix} = 12 \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = 60$$

$$R \rightarrow \frac{1}{2} |\vec{a} \times \vec{b}| = 20$$

$$\frac{1}{2} |(2\vec{a} + 3\vec{b}) \times (\vec{a} - \vec{b})|$$

$$\frac{1}{2} |-2(\vec{a} \times \vec{b}) - 3(\vec{a} \times \vec{b})|$$

$$\frac{5}{2} \times 40 = 100$$

$$S \rightarrow |\vec{a} \times \vec{b}| = 30$$

$$\Rightarrow |(\vec{a} + \vec{b}) \times \vec{a}| = |\vec{b} \times \vec{a}| = 30$$

VECTOR ALGEBRA

JEE ADVANCED - VOL - I

125. (A) $(\vec{b} - \vec{d}) \times \vec{c} = 8\vec{i} + 4\vec{j} - 4\vec{k}$

$\vec{a} + \vec{c} = 4\vec{i} + 2\vec{k}$

So, projection $= \frac{32 + 0 - 8}{\sqrt{64 + 16 + 16}} = \sqrt{6}$

(B) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{1}{2}(1 + 1 + 1) = -\frac{3}{2}$

(C) $[\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}] = [\vec{a} \vec{b} \vec{c}]^2$

(D) $c(\log_2 x)^2 - 12 + 6c \log_2 x < 0$
 $\Rightarrow c < 0$

and $36c^2 + 48c < 0 \Rightarrow c \in \left(-\frac{4}{3}, 0\right)$

126. (A) The point $(1+t, 1, -t)$ where t is real parameter, always lie on the given line.

(B) Solving the two equations we have

$(1+t)2 + 1 - t = 2$

$\Rightarrow t = -1$

So ; $\vec{r} = \hat{j} + \hat{k}$

(C) Any point on the line is $(1+t, 1-t)$

$\sqrt{(1+t-1)^2 + 1+t^2} = 3$

$\Rightarrow t = \pm 2$

So, the point is $3\vec{i} + \vec{j} - 2\vec{k}$ and

$-\vec{i} + \vec{j} + 2\vec{k}$

(D) Let $\vec{C} = x\hat{i} + y\hat{j} + z\hat{k}$

$\begin{vmatrix} x & y & z \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = 4$

$\Rightarrow |(y-z) + 1(x-2y)| = 4$

$\Rightarrow |x - y - z| = 4$

127. [A] $\cos \theta = \frac{(\vec{i} + 2\vec{j} + 3\vec{k}) \cdot (3\vec{i} + 2\vec{j} + 6\vec{k})}{3 \times 7} = \frac{19}{21}$

[B] perpendicular vector

$= (\vec{i} - 2\vec{j} - \vec{k}) \times (3\vec{i} - 2\vec{j} - \vec{k}) = -2\vec{j} + 4\vec{k}$

Angle θ with $\vec{i} + \vec{j} + \vec{k}$ is given by

$\cos \theta = \frac{(-2\vec{j} + 4\vec{k}) \cdot (\vec{i} + \vec{j} + \vec{k})}{|-2\vec{j} + 4\vec{k}| |\vec{i} + \vec{j} + \vec{k}|} = \frac{1}{\sqrt{15}}$

[C] $\cos \theta = \frac{(\vec{i} + \vec{j} + \vec{k}) \cdot (\vec{i} + \vec{j})}{\sqrt{3}\sqrt{2}} = \sqrt{\frac{2}{3}}$

[D] Since $\vec{a}$ and $\vec{b}$ are linearly independent

$\therefore [\vec{u} \vec{v} \vec{w}] = [\vec{u} \vec{v} \vec{z}] = 0$

$\vec{u}, \vec{v}, \vec{w}$ and $\vec{z}$ are coplanar

$\Rightarrow \vec{v} \times \vec{w}$ is perpendicular to

INTEGER QUESTIONS

128. $\overline{AA_1} = \frac{\overline{AB} + \overline{AC}}{2}$

$\overline{AB_1} = \frac{\overline{AC} + \overline{AD}}{2}$

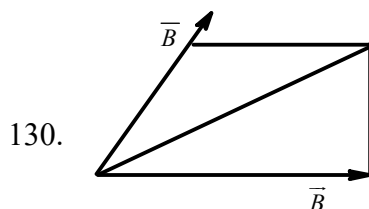
$\therefore \overline{AA_1} + \overline{AB_1} = \frac{3}{2} \overline{AC}$

129. $9PG^2 + (\overline{AB})^2 + (\overline{AC})^2 + (\overline{BC})^2$

$= 3 \left[(\overline{PA})^2 + (\overline{PB})^2 + (\overline{PC})^2 \right]$

$= 36 + (a^2 + b^2 + c^2) = 3(1 + 9 + 4)$

$a^2 + b^2 + c^2 = 42 - 36 = 6$



130.

Vector bisecting internal angle between $\vec{A}$ and

$\vec{B} = \frac{\vec{A}}{|\vec{A}|} + \frac{\vec{B}}{|\vec{B}|} = \vec{A} + \vec{B} \Rightarrow \alpha = 1$

$$131. \vec{p} \cdot \vec{q} = \frac{2\sqrt{2}}{3} \cdot 1 \cdot \frac{1}{\sqrt{2}} = \frac{2}{3}$$

$$\vec{a} + \vec{b} = 6\vec{p} - \vec{q}$$

$$|\vec{a} + \vec{b}|^2 = 36|\vec{p}|^2 + |\vec{q}|^2 - 12\vec{p} \cdot \vec{q}$$

$$36 \cdot \frac{8}{9} + 1 - 12 \cdot \frac{2}{3} = 25$$

$$\vec{a} - \vec{b} = -(4\vec{p} + 5\vec{q})$$

$$|\vec{a} - \vec{b}|^2 = 16|\vec{p}|^2 + 25|\vec{q}|^2 + 40\vec{p} \cdot \vec{q}$$

$$= 16 \cdot \frac{8}{9} + 25 + 40 \cdot \frac{2}{3} > 25$$

$\therefore \vec{a} + \vec{b}$ is the shorter diagonal with length 5.

$$132. \vec{OP} = x_1\hat{i} + y_1\hat{j}$$

$$\vec{OQ} = x_2\hat{i} + y_2\hat{j}$$

$$\vec{OP} \cdot \hat{i} = x_1$$

$$\vec{OQ} \cdot \hat{i} = x_2$$

$$x_1 = -1$$

$$x_2 = 2$$

$$y_1 = 2, y_2 = 1$$

$$\vec{OQ} = -2\hat{i} + \hat{j}$$

$$\vec{OP} = -\hat{i} + 2\hat{j}$$

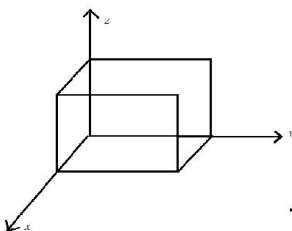
$$|\vec{OQ}| = \sqrt{4+1} = \sqrt{5} \quad |\vec{OP}| = \sqrt{1+4} = \sqrt{5}$$

$$133. \cos \theta = \frac{-a^2 - b^2 + c^2}{a^2 + b^2 + c^2}$$

$$\cos \phi = \frac{-a^2 - b^2 - c^2}{a^2 + b^2 + c^2}$$

$$\cos \psi = \frac{-a^2 - b^2 - c^2}{a^2 + b^2 + c^2}$$

$$|\cos \theta + \cos \phi + \cos \psi| = 1$$



134.

$$\vec{a} \cdot \vec{b} = [x\hat{i} + (x+1)\hat{j} + \hat{k}] \cdot [(x+1)\hat{i} + \hat{j} + a\hat{k}]$$

$$= x(x+1) + x - 1 + a$$

$$= x^2 + 2x + a - 1$$

We must have $\vec{a} \cdot \vec{b} > 0, \forall x \in R$

$$\Rightarrow x^2 + 2x + a - 1 > 0, \forall x \in R$$

$$= 4 - 4(a - 1) < 0 \Rightarrow a > 2$$

$$135. \vec{n} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}, \text{ where } a_1^2 + a_2^2 + a_3^2 = 1$$

Given that $\vec{u} \cdot \vec{n} = 0 \Rightarrow a_1 + a_2 = 0$

$$a_1 = a_2 = 0$$

and $a_3 = 1 \text{ or } -1$

$$136. 6 - 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} - 2\vec{c} \cdot \vec{a} = 9$$

$$(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = \frac{-3}{2}$$

$$|\vec{a} + \vec{b} + \vec{c}|^2 \geq 0$$

$$3 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq 0$$

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} \geq \frac{-3}{2}$$

$$\text{Since } \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \frac{-3}{2}$$

$$\Rightarrow |\vec{a} + \vec{b} + \vec{c}| = 0 \Rightarrow \vec{a} + \vec{b} + \vec{c} = 0$$

$$\Rightarrow |2\vec{a} + 5(-\vec{a})| = |3\vec{a}| = 3$$

$$137. \text{ We have } |\vec{a} + \vec{b}| = \sqrt{3}$$

$$\Rightarrow |\vec{a} + \vec{b}|^2 = 3$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b}) = 3$$

$$\Rightarrow 1 + 1 + 2(\vec{a} \cdot \vec{b}) = 3$$

$$\Rightarrow \vec{a} \cdot \vec{b} = \frac{5}{2}$$

$$\text{Now } \vec{c} - \vec{a} - 2\vec{b} = 3(\vec{a} \times \vec{b})$$

$$\Rightarrow (\vec{c} - \vec{a} - 2\vec{b}) \cdot \vec{b} = 3[(\vec{a} \times \vec{b}) \cdot \vec{b}]$$

$$\Rightarrow \vec{c} \cdot \vec{b} - \vec{a} \cdot \vec{b} - 2(\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

$$(\therefore [\vec{a} \vec{b} \vec{b}] = 0)$$

$$\Rightarrow \vec{c} \cdot \vec{b} - \frac{1}{2} - 2 \times 1 = 0$$

$$\Rightarrow \vec{c} \cdot \vec{b} = \frac{5}{2}$$

138

$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOC} = \frac{(1/2)|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}{(1/2)|\vec{a} \times \vec{b}|}$$

$$\text{Now } \vec{a} + 2\vec{b} + 3\vec{c} = 0$$

$$\text{cross with } \vec{b}, \vec{a} \times \vec{b} + 3\vec{c} \times \vec{b} = 0$$

$$\Rightarrow \vec{a}, 2\vec{a} \times \vec{b} + 3\vec{a} \times \vec{c} = 0$$

$$\Rightarrow \vec{a} \times \vec{b} = 3(\vec{b} \times \vec{c})$$

$$\text{Hence, } \vec{a} \times \vec{b} = \frac{3}{2}(\vec{c} \times \vec{a}) = 3(\vec{b} \times \vec{c})$$

$$\text{Let } (\vec{c} \times \vec{a}) = \vec{p}. \text{ Then}$$

$$\vec{a} \times \vec{b} = \frac{3\vec{p}}{2} \quad \vec{b} \times \vec{c} = \frac{\vec{p}}{2}$$

Hence, the ratio is

$$= \frac{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}{|\vec{p}|}$$

$$= \frac{|(3\vec{p}/2) + (\vec{p}/2) + \vec{p}|}{|\vec{p}|}$$

$$= \frac{3|\vec{p}|}{|\vec{p}|} = 3$$

$$139. \text{ Given } (n-1) \sin \frac{2\pi}{n} = (n-1) \cos \frac{2\pi}{n}$$

$$\Rightarrow n = 8$$

$$140. [\overline{AB} \quad \overline{AC} \quad \overline{AD}] = 0$$

$$141. 2[\vec{a}\vec{b}\vec{c}] + [\vec{a}\vec{b}\vec{c}]^2 + 0$$

$$= 2 \times 1 + 1^2 = 3$$

142. The three adjacent sides of tetrahedron is given by

$$\frac{(\hat{i} + \hat{j}) \times (\hat{j} + \hat{k})}{|(\hat{i} + \hat{j}) \times (\hat{j} + \hat{k})|}, \frac{(\hat{j} + \hat{k}) \times (\hat{k} + \hat{i})}{|(\hat{j} + \hat{k}) \times (\hat{k} + \hat{i})|}, \frac{(\hat{k} + \hat{i}) \times (\hat{i} + \hat{j})}{|(\hat{k} + \hat{i}) \times (\hat{i} + \hat{j})|}$$

$$\text{i.e. } \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}, \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}, \frac{-\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$$

$$V = \frac{1}{6} \times \frac{1}{3\sqrt{3}} \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix} = \frac{2}{9\sqrt{3}}$$

$$\text{so ; } 9\sqrt{3}V = 2$$

$$143. \ell(\vec{b} \times \vec{c}) + m(\vec{c} \times \vec{a}) + n(\vec{a} \times \vec{b}) = 4\hat{i}$$

taking dot product with $\vec{a}$ both sides $2\ell = 4\vec{a} \cdot \hat{i}$

similarly $2m = 4\vec{b} \cdot \hat{i}$ and $2n = 4\vec{c} \cdot \hat{i}$

on adding $2(\ell + m + n) = 4(\vec{a} + \vec{b} + \vec{c}) \cdot \hat{i}$

$$= 4(3\hat{i} \cdot \hat{i}) = 12$$

$$\therefore \ell + m + n = 6$$

$$144. \text{ volume of the parallelopiped } = |[\vec{a}\vec{b}\vec{c}]|$$

$$\Rightarrow (\text{area of the base parallelogram}) h = |[\vec{a}\vec{b}\vec{c}]|$$

$$\Rightarrow |\vec{a} \times \vec{b}| h = |[\vec{a}\vec{b}\vec{c}]| \Rightarrow \sqrt{38}h = 4$$

$$\Rightarrow h = \frac{4}{\sqrt{38}}$$

$$38h^2 = 4 \times 4$$

$$\Rightarrow 19h^2 = 8$$

145. The foot of the point (1,3,4) in the plane $2x - y + z + 3 = 0$ is given by

$$\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = \frac{1}{6}(2-3+4+3) = -1$$

$\therefore F = (-1, 4, 3)$ Whose distance from the x-axis

$$\text{is } \sqrt{4^2 + 3^2} = 5$$

146. The image of the point (2, -3, 3) in the plane $x - 2y - z + 1 = 0$ is given by

$$\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z-3}{-1} = \frac{-2}{6}(2+6-3+1) = -2$$

$\therefore P(0, 1, 5)$ whose distance from the z-axis is $\sqrt{0^2 + 1^2} = 1$

147. $(\vec{a} + \vec{b}) \cdot \vec{c} = 0 \Rightarrow \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c} = 0$

$$(\vec{a} + \vec{b}) \cdot \vec{a} = 0 \Rightarrow \vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{c} = 0 \quad \text{Therefore,}$$

$$2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0 \quad \text{Now,}$$

$$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 50$$

$$|\vec{a} + \vec{b} + \vec{c}| = 5\sqrt{2}$$

148. Taking O as the origin, let the position vectors of A, B and C be $\vec{a}, \vec{b}$ and $\vec{c}$, respectively. Then the position vectors of G_1, G_2 and G_3 are

$$\frac{\vec{b} + \vec{c}}{3}, \frac{\vec{c} + \vec{a}}{3} \text{ and } \frac{\vec{a} + \vec{b}}{3}$$

$$V_1 = \frac{1}{6}[\vec{a} \vec{b} \vec{c}] \text{ and } V_2 = [\vec{OG}_1 \vec{OG}_2 \vec{OG}_3]$$

$$\text{Now } V_2 = [\vec{OG}_1 \vec{OG}_2 \vec{OG}_3]$$

$$\Rightarrow V_2 = \frac{1}{27}[\vec{b} + \vec{c} \vec{c} + \vec{a} \vec{a} + \vec{b}]$$

$$\Rightarrow V_2 = \frac{2}{27}[\vec{a} \vec{b} \vec{c}]$$

$$\Rightarrow V_2 = \frac{2}{27} \times 6V_1 \Rightarrow 9V_2 = 4V_1.$$

149. We know that $[\vec{e}_1 \vec{e}_2 \vec{e}_3][\vec{E}_1 \vec{E}_2 \vec{E}_3]$

$$= \begin{vmatrix} \vec{e}_1 \cdot \vec{E}_1 & \vec{e}_1 \cdot \vec{E}_2 & \vec{e}_1 \cdot \vec{E}_3 \\ \vec{e}_2 \cdot \vec{E}_1 & \vec{e}_2 \cdot \vec{E}_2 & \vec{e}_2 \cdot \vec{E}_3 \\ \vec{e}_3 \cdot \vec{E}_1 & \vec{e}_3 \cdot \vec{E}_2 & \vec{e}_3 \cdot \vec{E}_3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

$$150. \left| (\vec{a} \times \vec{b}) \times \vec{c} \right| = |\vec{a} \times \vec{b}| |\vec{c}| \sin 30^\circ$$

$$= \frac{1}{2} |\vec{a} \times \vec{b}| |\vec{c}|$$

We have $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$

$$\Rightarrow \vec{a} \times \vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$\Rightarrow |\vec{a} \times \vec{b}| = \sqrt{9} = 3$$

Also given $|\vec{c} - \vec{a}| = 2\sqrt{2}$

$$\Rightarrow |\vec{c} - \vec{a}|^2 = 8$$

$$\Rightarrow |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{a} \cdot \vec{c} = 8$$

Given $|\vec{a}| = 3$ and $\vec{a} \cdot \vec{c} = |\vec{c}|$.

Therefore, $|\vec{c}|^2 - 2|\vec{c}| + 1 = 0$

$$\Rightarrow (|\vec{c}| - 1)^2 = 0 \Rightarrow |\vec{c}| = 1$$

Substituting the values of $|\vec{a} \times \vec{b}|$ and $|\vec{c}|$ in (1),

we get

$$\left| (\vec{a} \times \vec{b}) \times \vec{c} \right| = \frac{1}{2} \times 3 \times 1 = \frac{3}{2}$$

151. Given that $\vec{V} = 2\vec{i} + \vec{j} - \vec{k}$ and $\vec{W} = \vec{i} + 3\vec{k}$ and $\vec{u}$ is a unit vector. Thus $|\vec{u}| = 1$.

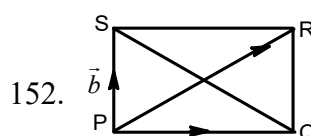
Now $[\vec{U} \vec{V} \vec{W}]$

$$= \vec{U} \cdot (\vec{V} \times \vec{W}) = \vec{U} \cdot ((2\hat{i} + \hat{j} - \hat{k}) \times (\hat{i} + 3\hat{k}))$$

$$= \vec{U} \cdot (3\hat{i} - 7\hat{j} - \hat{k})$$

$$= \sqrt{3^2 + 7^2 + 1^2} \cos \theta, \text{ which is maximum when } \cos \theta = 1.$$

Hence, the maximum value of $[\vec{U} \vec{V} \vec{W}]$ is $\sqrt{59}$



VECTOR ALGEBRA

$$\vec{a} + \vec{b} = 3\hat{i} + \hat{j} - 2\hat{k}$$

$$\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{a} + \vec{b} \times \vec{a} - \vec{b} = -10\hat{i} + 10\hat{j} - 10\hat{k}$$

$$-2\vec{a} \times \vec{b} = -10\hat{i} + 10\hat{j} - 10\hat{k}$$

$$\vec{a} \times \vec{b} = -5(\hat{i} - \hat{j} + \hat{k})$$

$$\vec{c} \cdot \vec{a} \times \vec{b} = 10$$

$$\therefore k - 1 = 9$$

$$153. \quad 0 = \frac{p}{2} + q + \frac{r}{2}$$

$$[\overline{abc}] = \frac{p}{2} + \frac{q}{2} + r$$

$$[\overline{abc}] \therefore p = -q = r \Rightarrow \frac{p^2 + 2q^2 + r^2}{q^2} = 4$$

$$154. \quad a = |\overline{BC}|, b = |\overline{CA}|, c = |\overline{AB}|$$

$$\text{and } \overline{OG} = \frac{\overline{OA} + \overline{OB} + \overline{OC}}{3}$$